

Supplementary Material: Technical Report for 4343MMEBX

Fusing Geographic Context with Visual Features for Cryptic Species Identification

Group 10: Shéylle Green, Tara Splint, Luca Pouw, Marc Wijnands

SUPPLEMENTARY MATERIAL

This supplementary document provides additional experimental detail, full result tables, and diagnostic figures supporting the main report. All models were trained on CrypticBio-Common (158 species, 15,801 observations, 70/15/15 train/val/test split, seed 42) using a ResNet-50 backbone pre-trained on ImageNet, with AdamW optimisation, cosine annealing, early stopping (patience 5), label smoothing ($\epsilon=0.1$), batch size 32, and backbone learning rate 3×10^{-5} with heads and encoders at 10^{-3} . Continuous geographic encodings (Raw, Wrap, SH) were projected to a fixed 64-d representation via a two-layer MLP before fusion; discrete encodings (Hex, Geo-label) used an internal projection to 128-d. All 14 multimodal configurations share identical image preprocessing and classifier architecture, ensuring that performance differences are attributable solely to the geographic encoding or fusion strategy.

Supplementary Figure 1 — Full model performance

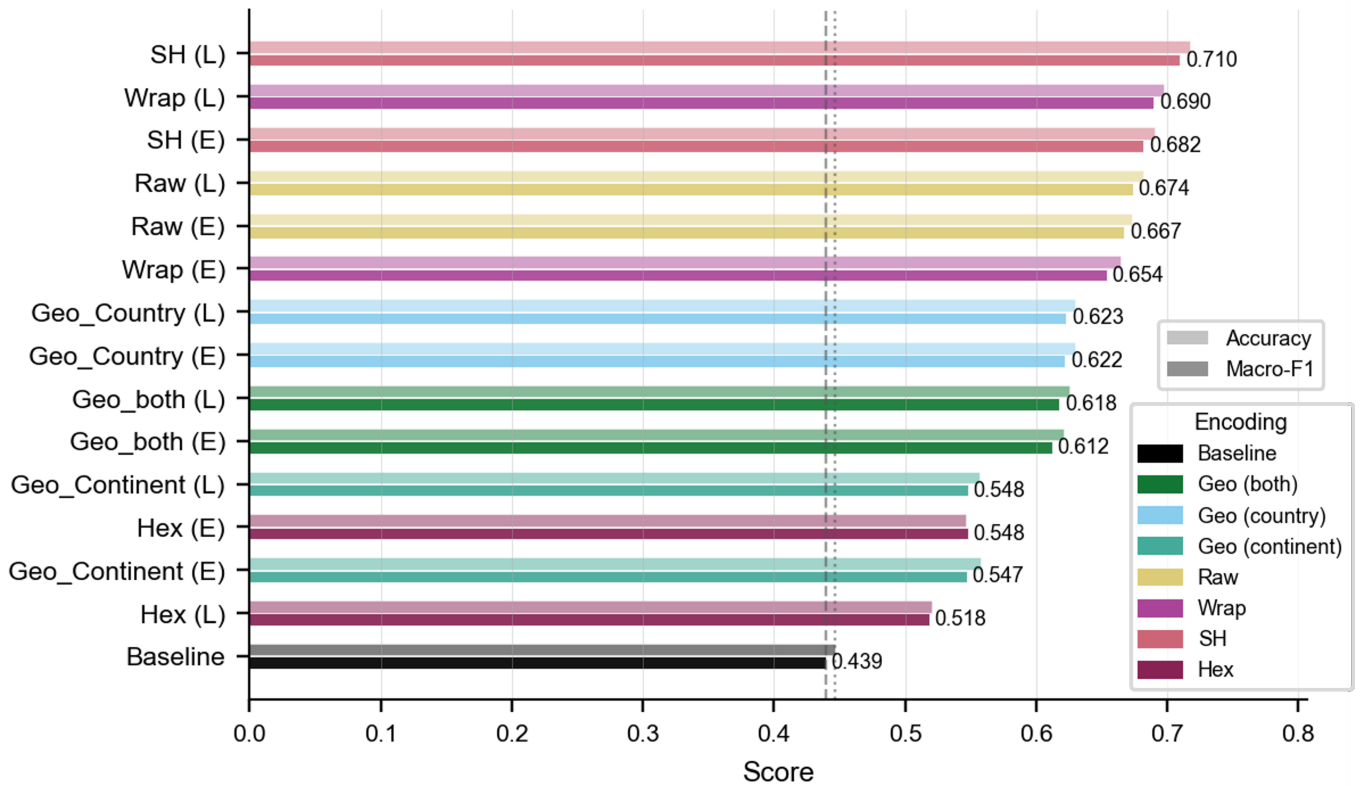


Figure 1. Top-1 accuracy (transparent bar) and macro-F1 (solid bar) for all 15 configurations (14 multimodal + baseline), sorted by macro-F1. The dashed vertical line marks the image-only baseline (macro-F1 = 0.439). Bar colour denotes encoding family. SH(L) achieves the highest score on both metrics; Hex(L) is the only configuration that falls below Geo_Continent on macro-F1. Within each continuous encoding family, late fusion consistently outperforms early fusion.

Supplementary Figure 2 — Range breadth vs. F1 gain

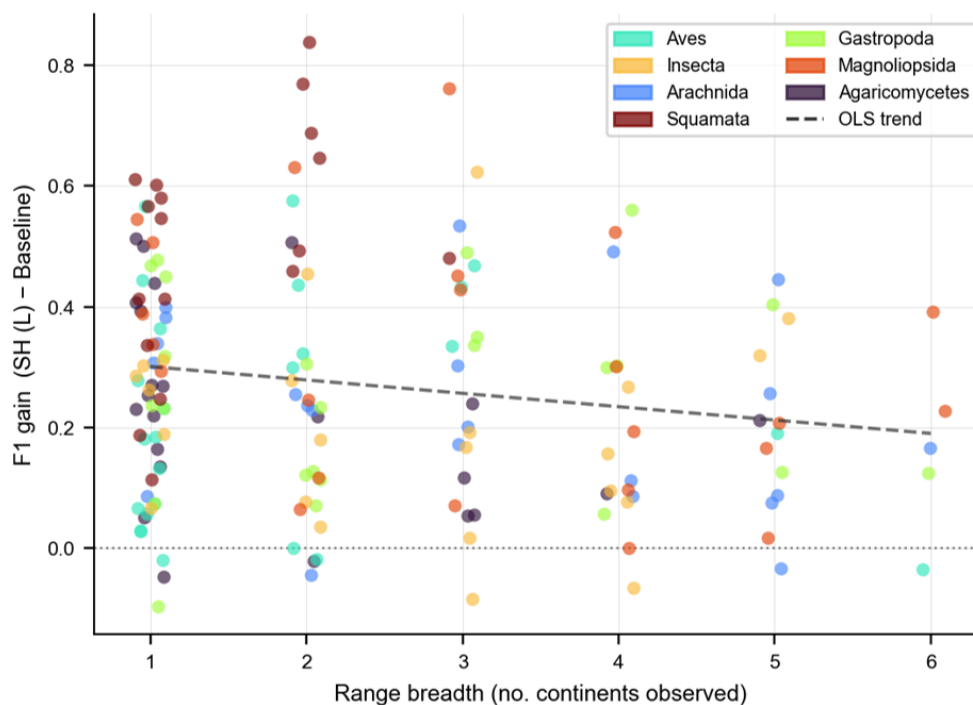


Figure 2. Per-species F1 gain (SH(L) – Baseline) plotted against range breadth, defined as the number of distinct continents in which a species was observed in the occurrence dataset. The dashed line is a linear OLS (ordinary least-squares) trend fitted across all 158 species. The mildly negative slope indicates that species observed across more continents do not gain more from geographic fusion; the discriminating signal operates through between-species range separation rather than within-species occurrence breadth. High within-group variance at each range breadth value confirms that continent count alone is not a reliable predictor of benefit. Colour denotes taxonomic class; most species are observed on 1–2 continents, reflecting the predominantly regional distribution of cryptic taxa.

Supplementary Figure 3 — Model × Taxon accuracy heatmap

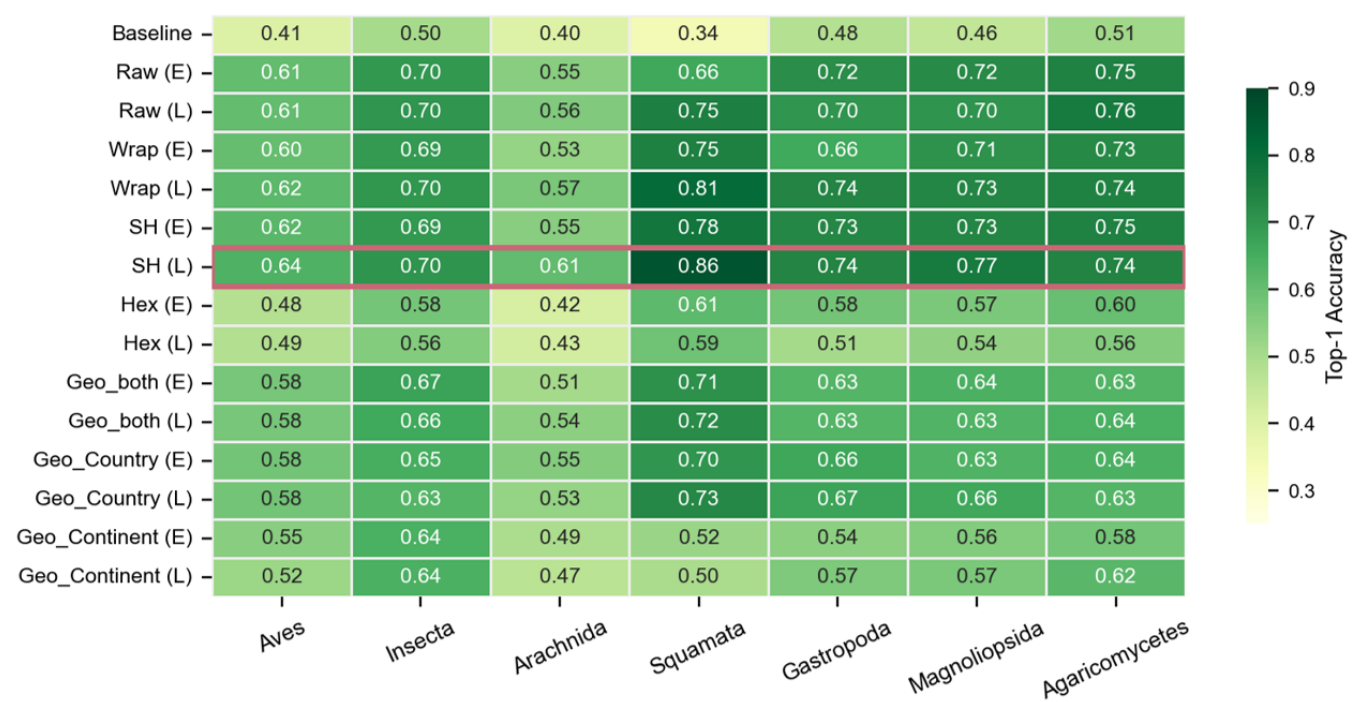


Figure 3. Top-1 accuracy for each model–taxon combination. Rows are models ordered by overall accuracy; columns are the seven taxonomic classes. The blue border highlights SH(L), the best-performing model. Squamata shows the largest absolute gain under SH(L) (0.34 baseline → 0.86), consistent with the strong allopatric range structure of cryptic reptile taxa. Hex encodings are consistently the weakest across every taxon, confirming that their underperformance is structural and not taxon-specific. Geo-label models perform competitively on taxa with strong continental-scale range separation (e.g. Squamata) but underperform on taxa with within-continent cryptic pairs.

Supplementary Table 1 — Top-20 misclassification pairs

Table 1. Top-20 most frequent misclassification pairs under the image-only baseline and corresponding counts under SH(L). Δ = Baseline – SH(L) (positive = improvement). ✓ = confirmed cryptic pair; — = non-cryptic pair. 18 of 20 pairs are confirmed cryptic; 14 are reduced to zero under SH(L). The two most resistant pairs (*pluchei*→*phalangioides* and *vivipara*→*agilis*) involve cryptic congeners with substantially overlapping European ranges.

True	Predicted	Taxon	Cryptic	Baseline	SH(L)	Δ
<i>Chilocorus cacti</i>	<i>C. renipustulatus</i>	Insecta	✓	10	0	+10
<i>Poecile carolinensis</i>	<i>P. atricapillus</i>	Aves	✓	8	1	+7
<i>Lacerta viridis</i>	<i>L. bilineata</i>	Squamata	✓	7	0	+7
<i>Holocnemus pluchei</i>	<i>Pholcus phalangioides</i>	Arachnida	✓	7	6	+1
<i>Monacha cantiana</i>	<i>Bradybaena similaris</i>	Gastropoda	✓	6	0	+6
<i>Zootoca vivipara</i>	<i>Lacerta agilis</i>	Squamata	✓	5	3	+2
<i>Steatoda triangulosa</i>	<i>S. grossa</i>	Arachnida	✓	5	0	+5
<i>Poecile atricapillus</i>	<i>P. carolinensis</i>	Aves	✓	5	0	+5
<i>Amanita citrina</i>	<i>A. cokeri</i>	Agaricomycetes	✓	5	0	+5
<i>Amanita pantherinoides</i>	<i>A. regalis</i>	Agaricomycetes	✓	5	0	+5
<i>Erigeron coulteri</i>	<i>Brachyscome decipiens</i>	Magnoliopsida	—	5	0	+5
<i>Erigeron annuus</i>	<i>E. strigosus</i>	Magnoliopsida	✓	5	3	+2
<i>Cantareus apertus</i>	<i>Zachrysia provisoria</i>	Gastropoda	✓	5	0	+5
<i>Steatoda nobilis</i>	<i>S. grossa</i>	Arachnida	✓	4	2	+2
<i>Gallotia galloti</i>	<i>Teira dugesii</i>	Squamata	—	4	0	+4
<i>Tripleurospermum inodorum</i>	<i>Matricaria chamomilla</i>	Magnoliopsida	✓	4	1	+3
<i>Eobania vermiculata</i>	<i>Otala lactea</i>	Gastropoda	✓	4	0	+4
<i>Cycloneda polita</i>	<i>C. sanguinea</i>	Insecta	✓	4	2	+2
<i>Chilocorus renipustulatus</i>	<i>C. cacti</i>	Insecta	✓	4	0	+4
<i>Cepaea nemoralis</i>	<i>C. hortensis</i>	Gastropoda	✓	4	2	+2

Supplementary Table 2 — Graph analysis results

Table 2. Graph-structural metrics of the weighted directed misclassification graph $G(V, E, W)$ per model, where edge (u, v) encodes the frequency of predicting species v when the ground truth is u . *Solved species*: species for which all predictions are correct (only self-loop edges). *Strong cliques*: mutually confusable species clusters (lower = better). *Strong components*: strongly connected components (higher = better). *Weak components*: weakly connected components; the dataset’s cryptic group structure defines 13 as a structural reference (higher = better). *Density*: fraction of possible directed edges present (lower = better). SH(L) achieves the most strongly connected components (25) and lowest density (0.027), confirming structural resolution of confusion clusters.

Model	Solved sp.	Cliques	Clique edges	Clique wt.	Max clique	Strong comp.	Weak comp.	Density	Correct pred.
Baseline	0	4	16	31	3	1	1	0.044	1057
Hex (L)	3	13	42	69	3	9	4	0.039	1231
Hex (E)	1	9	26	41	3	10	2	0.038	1294
Geo cont. (L)	0	5	10	15	2	8	1	0.035	1317
Geo cont. (E)	0	7	14	21	2	13	1	0.035	1319
Geo both (E)	1	8	20	37	3	17	2	0.031	1470
Geo both (L)	1	9	22	28	3	15	2	0.032	1481
Geo ctry. (E)	1	11	30	47	3	9	2	0.031	1490
Geo ctry. (L)	1	6	16	23	3	15	2	0.031	1491
Wrap (E)	1	16	48	68	3	16	2	0.029	1572
Raw (E)	1	9	26	34	3	11	2	0.030	1592
Raw (L)	3	13	38	50	3	20	4	0.028	1614
SH (E)	1	16	40	53	3	14	2	0.028	1634
Wrap (L)	1	9	22	29	3	19	3	0.028	1651
SH (L)	2	10	32	40	3	25	4	0.027	1697

Supplementary Figure 4 — In-neighbourhood distribution

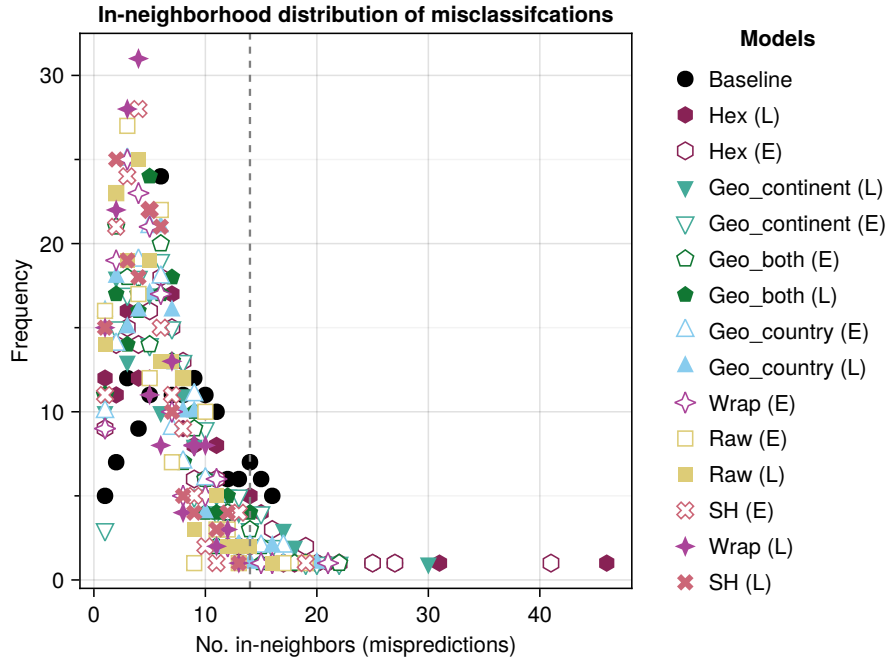


Figure 4. In-neighbourhood distribution of misclassified samples per model. The x-axis shows the number of distinct misclassified in-neighbours for each species — i.e. how many other species are incorrectly predicted as that species. The dotted vertical line marks the per-species test set size, beyond which a species receives more misclassified in-neighbours than it has true test samples, indicating it acts as a dominant attractor in the confusion network. Under the baseline and Hex encodings, at least one such attractor species exists (up to ~45 in-neighbours for Hex late), whereas continuous geo-fusion models substantially reduce in-neighbourhood counts, with SH(L) showing the most compressed distribution — consistent with its lowest graph density and highest number of strongly connected components.